What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the [Università dell'Insubria](https://www.uninsubria.eu/) in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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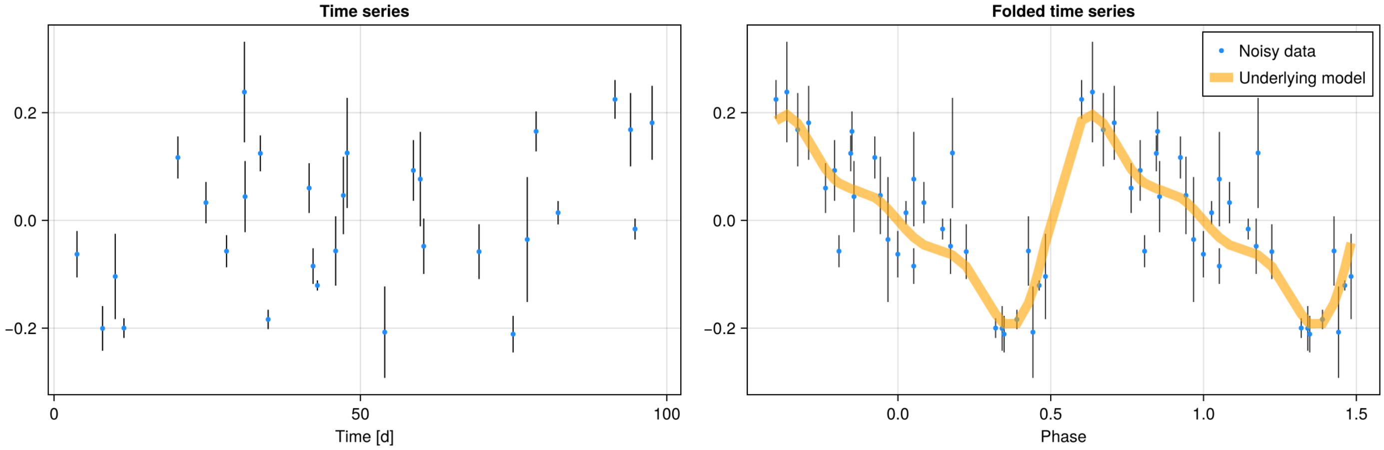
Exercise with non-parametric periodograms

- In this exercise, we are going to work with a simple irregulary sampled time series geberated using a harmonic model composed of three sine waves. The fundamental frequency is "2.0" and the observation lasts "100.0" inverse units of the frequency, with 30 points.

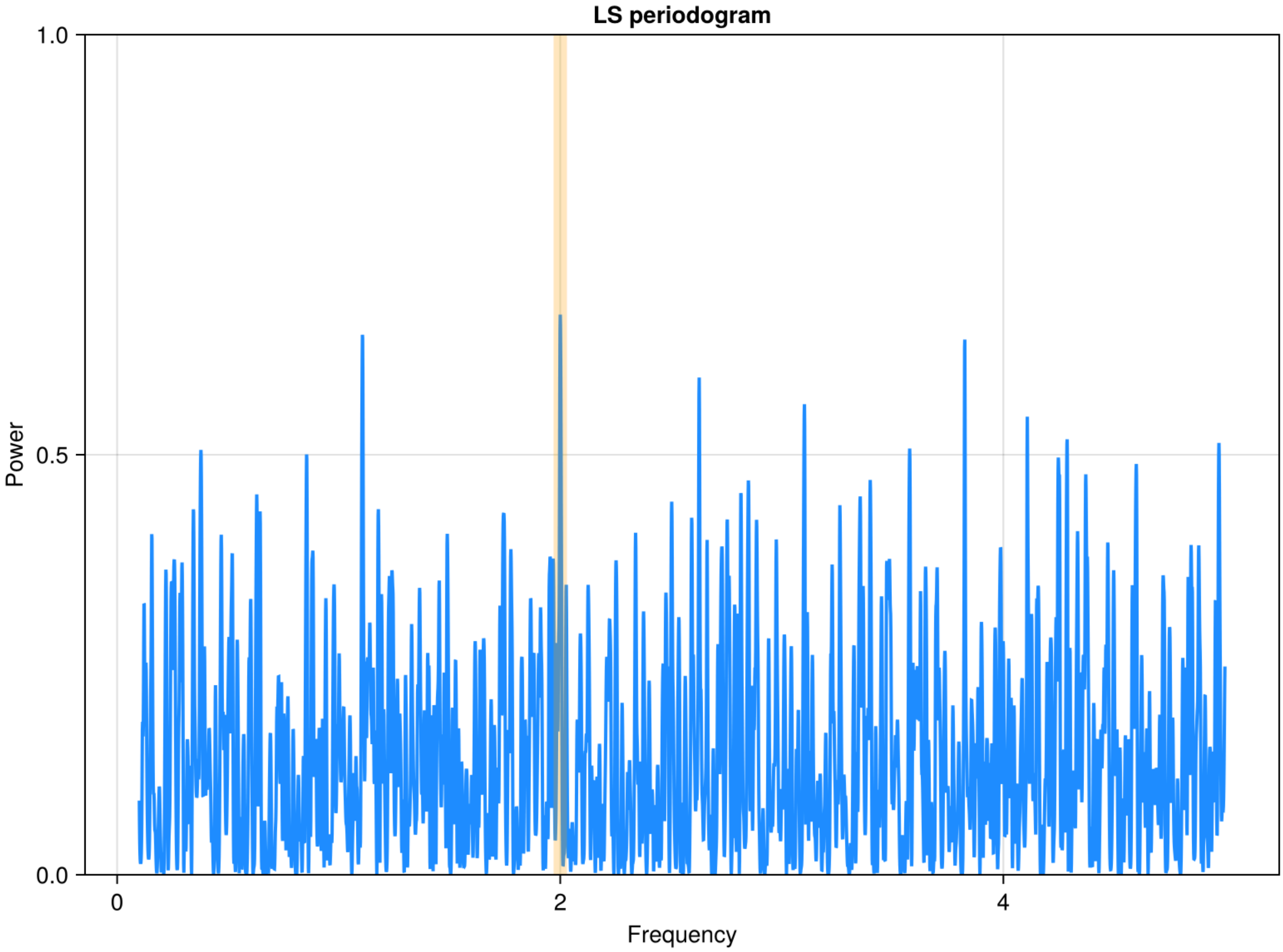


The simulated data are sufficiently close to a real dataset, composed by a modest number of heteroscedastic observations carried out when possible, i.e. irregularly sampled during the "source" monitoring.

Finding the best frequency/period using a Lomb-Scargle algorithm

- Before moving to non-parametric methods let's analyse our time-series with the *de facto* standard algorithm for irregularly sampled time-series: the Lomb-Scargle (LS) algorithm.
- Let's remind us that for evenly sampled light-curves the LS reduces to the plain discrete Fourier transform.

Best frequency: 2.00 corresponding to period: 0.50

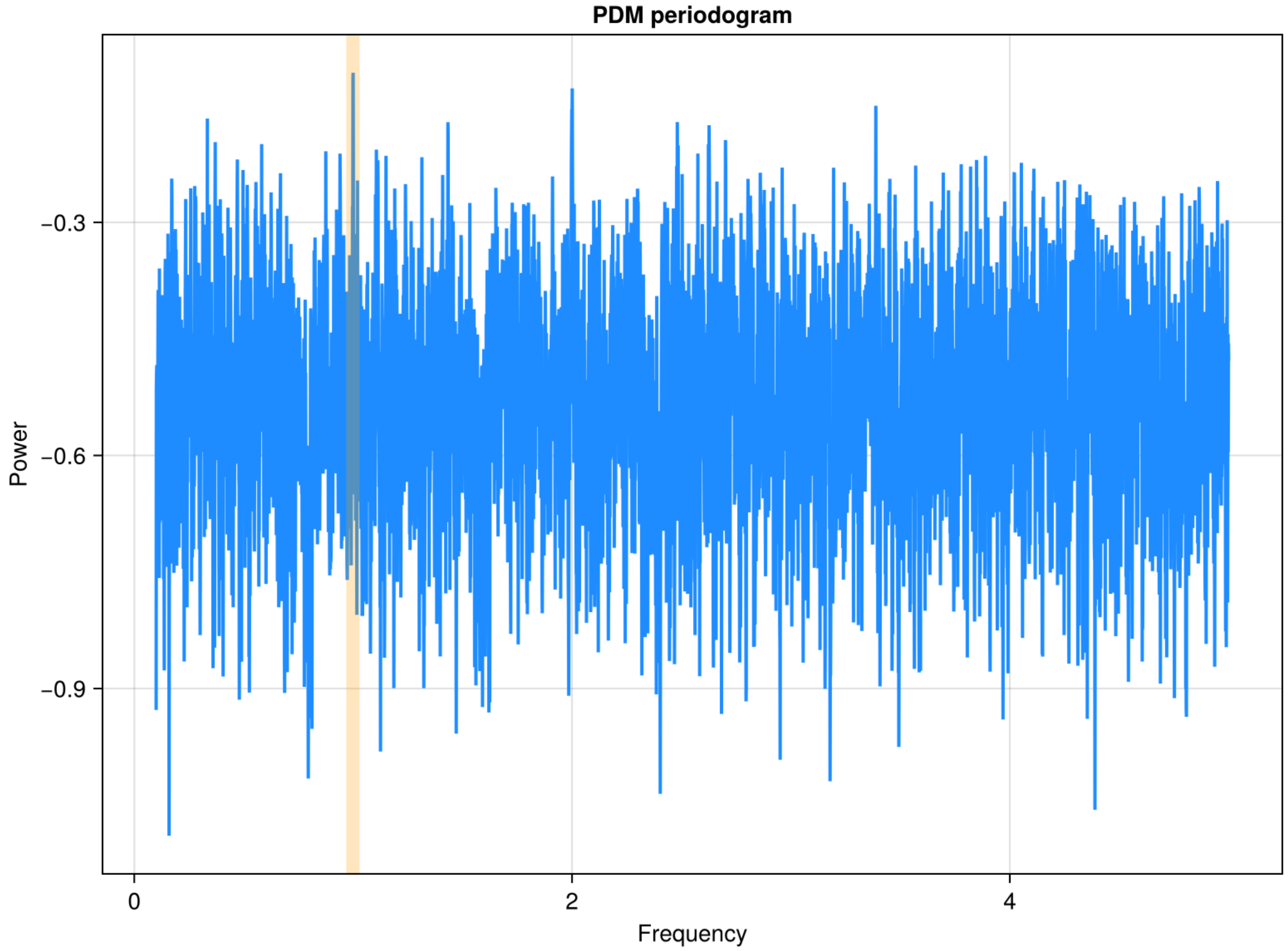


- We see immediately that the LS periodogram indeed does show a peak at the expected frequency.
    - Yet, even without a detailed statistical analysis, it is clear that its significance, compared to the several other peaks of similar power (the scale is linear) should likely be modest.
    - This is likely due to the relatively modest number of available observations, and the variability shape that is not truly sinusoidal. These two factors make hard for the algorithm to distiguish the *true* frequency from the harmonics.

The LS periodogram in this rather typical (although, by no means, trivial) case is difficult to interpret. We now move to non-parametric tools.

Phase dispersion minimization (PDM)

- First, we try to apply one of the most widely knwon parametric method: the PDM.
- Of course, let's assume that we do not know the best frequency for this time series.
- To find it, we sweep over a linear array of frequencies and find the ones that optimize the selected criterion.

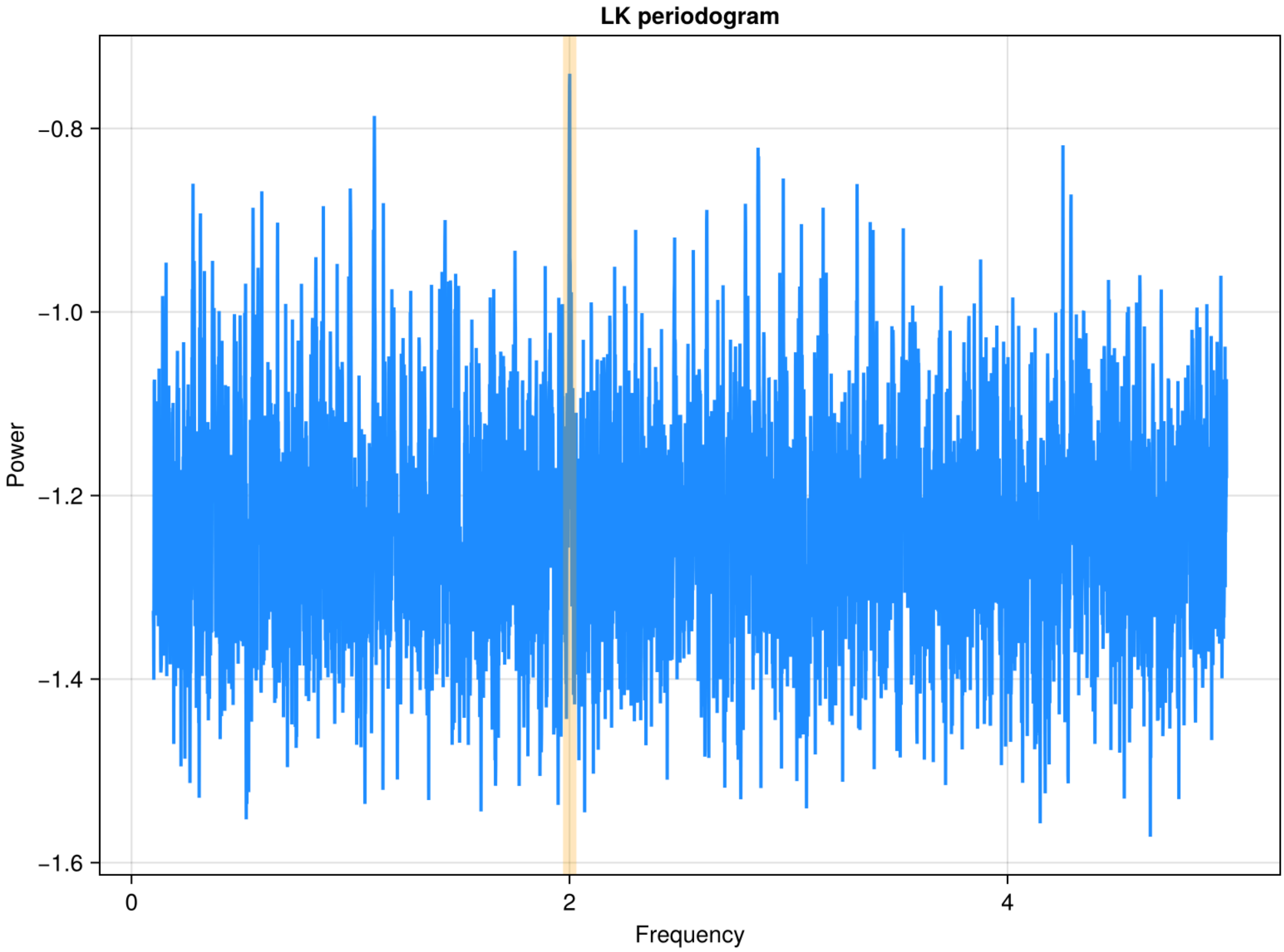


Best frequency: 1.00 corresponding to period: 1.00

- The PDM algorithm cannot recover the right input frequency as the highest peak in the periodogram.
- The situation is indeed analogous to what we saw for the LS periodogram. Several peaks of similar power are present making the interpretation of the periodogram difficult.

Lafler-Kinman's string length

- This is one of the possible variants of string-length methods

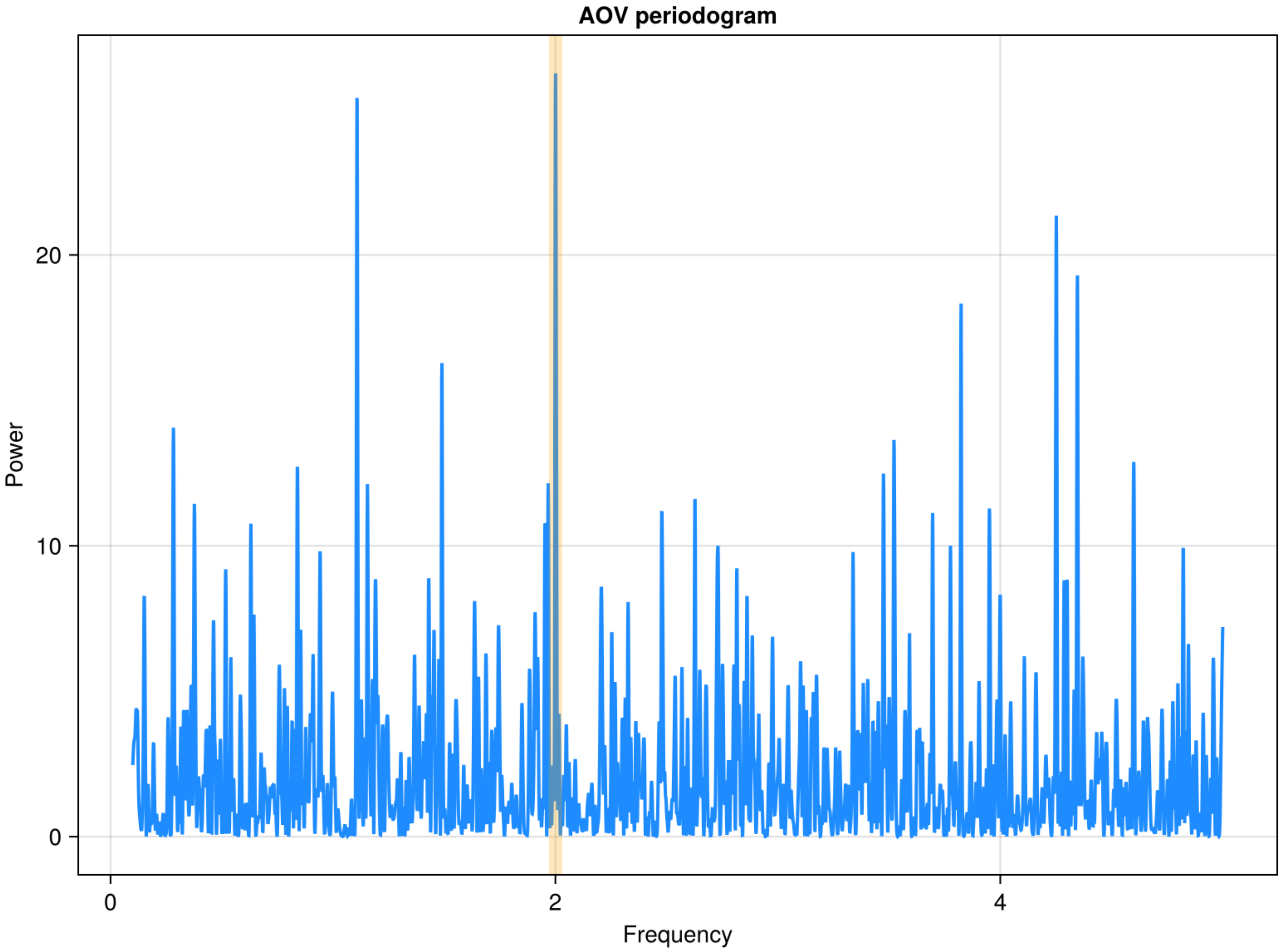


Best frequency: 2.00 corresponding to period: 0.50

- Here we have a situation similar to the LS periodogram. We can retrieve the right frequency but in a very noisy periodogram.

AoV analysis

- The AoV algorithm has been designed to model any light-curve shape with an approadh still close to the one adopted for the LS algorithm.

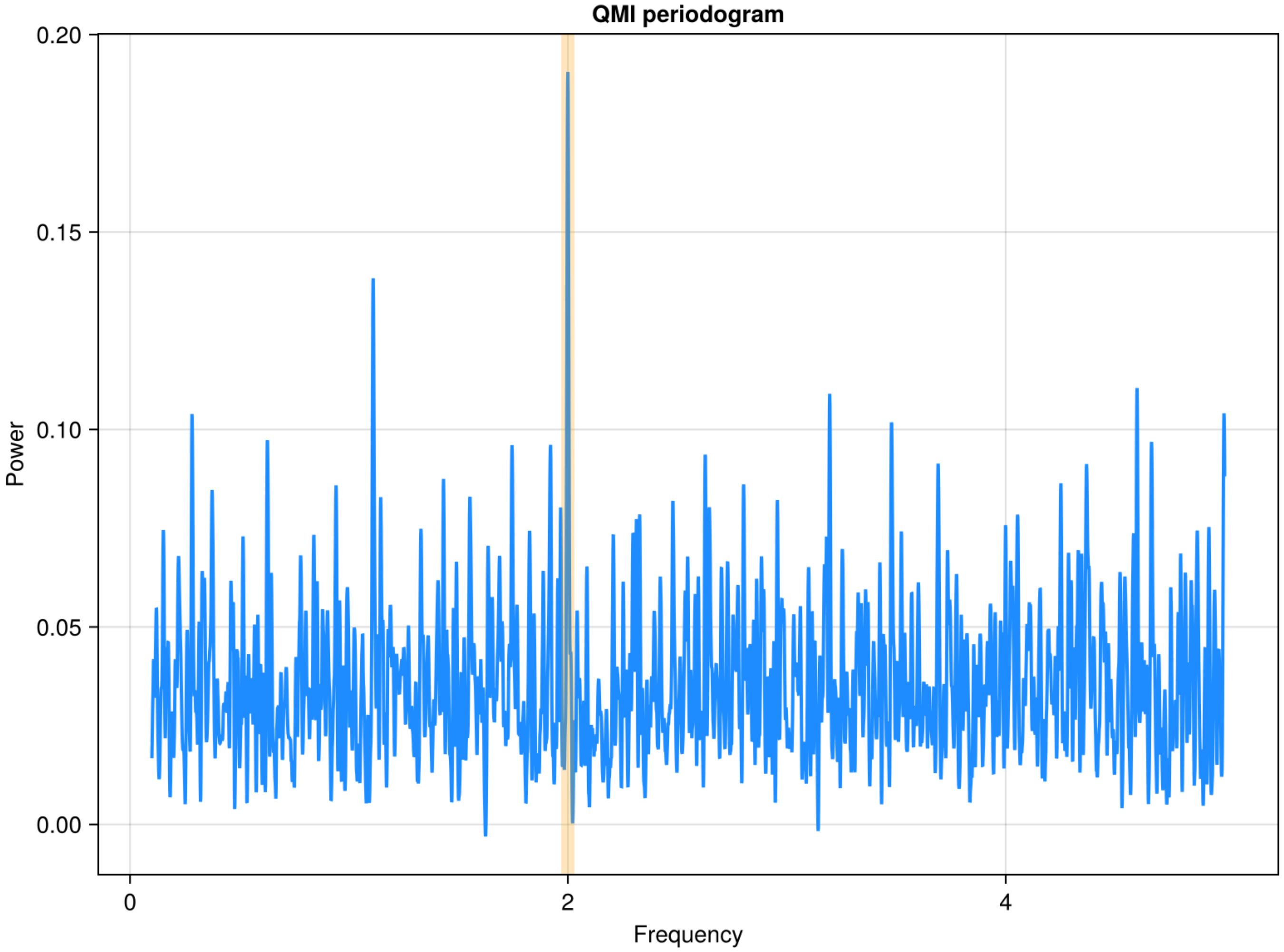


Best frequency: 2.00 corresponding to period: 0.50

- Another noisy periodogram, but again the right frequency is identified. As for most of the previous cases, the harmonic at half frequency is comparable in power to the *true* frequency using to generate the input data.

Quadratic Mutual Information

- And finally let's try with one of the algorithms based on the analysis of the probability densities of the input fluxes and phased data.



Best frequency: 2.00 corresponding to period: 0.50

- Indeed, the periodogram based on the QMI appears to be definitely the most convincing, with the right period standing above the alternatives.

Please, be aware that with a different dataset results can change, this simple exercize does not authorize, e.g., to judge the PDM technique as the least effective in general.

- It is indeed true that the QMI periodogram seems to be more effective than the other explored algorithms.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- Huijse et al. (2018) - Robust Period Estimation Using Mutual Information for Multiband Light Curves in the Synoptic Survey

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about non-parametric analysis	Science case about GRBs	Course Summary
html	Lecture about non-parametric analysis	Science case about GRBs	Course Summary

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